Phenix Installation and Setup

4 documentation pages
Generated from local Phenix documentation

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Creating a Windows installer for PHENIX

creating_windows_installer.html

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Introduction

Once you are satisfied that your Windows build of PHENIX passes all tests you may want to bundle it into an executable setup program to make installation easy for users. Currently the lightweight tool Makensis from Nullsoft Scriptable Install System is used for creating the installer.

Download and install it on your PC and modify the %PATH% environment variable to make makensis.exe accesible from a command prompt. It is useful to install the syntax highlightening editor "HM NIS Edit" for editing .nsi scripts. In addition it provides context sensitive help on keywords in the scripts.

How makensis.exe is run

Whether the build is done through Buildbot or done manually creating the installer starts with calling

```
build\bin\libtbx.create_installer.bat
```

with appropriate arguments as in the last line of example build script, Phenixbuild.cmd. It calls

```
cctbx_project\libtbx\command_line\create_installer.py
```

which calls

```
cctbx_project\libtbx\auto_build\create_installer.py
```

which calls

```
cctbx_project\libtbx\auto_build\create_windows_installer.py
```

which creates a short preamble of definitions such as version of Phenix and location of directories and then calls

```
Makensis.exe /OMakeWindowsInstaller.log /NOCD /V4 tmp\tmpinstscript.nsi
```

The file, tmpinstscript.nsi, is the script specifying which files to include and where to place them on the users PC. It is composed of the preamble prepended to the file

```
cctbx_project\libtbx\auto_build\mainphenixinstaller.nsi
```

which contains the bulk of information for creating the setup program. The instructions in this file are documented in the online help. Once Makensis has finished running the setup program is available as a file with the name:

```
dist\&lt;version&gt;\Phenix-&lt;version&gt;-&lt;platform&gt;-Setup.exe
```

which can then be deployed to users.

A few details of what is installed

The mainphenixinstaller.nsi script has been written to allow the user to install Phenix as a standard user as well as an administrator. The macro _UserIsAdmin is used for evaluating if the installer was invoked with admin rights. In the latter case the produced setup program should be invoked with administrative privileges. Phenix will then default to be installed in the directory pointed to by the %ProgramFiles(x86)% or %ProgramFiles% environment variables. The function .onInit is used for determining the location of the installation directory according to user rights. The installer will also allow the user to install the C/C++ source code used for the build. It will create a Startup group named Phenix\<version> which contains links to

documentation, the main GUI, a Phenix.python prompt, a command prompt where the Phenix environment variables have been defined, the uninstaller program.

- documentation,
- the main GUI,
- a Phenix.python prompt,
- a command prompt where the Phenix environment variables have been defined,
- the uninstaller program.

The installer will also create a registry value of type string in the key:

```
HKEY_CURRENT_USER\Software\Phenix\&lt;version&gt;\InstallPath
```

or, if the installer is invoked with administrative privileges:

```
HKEY_LOCAL_MACHINE\SOFTWARE\Phenix\&lt;version&gt;\InstallPath
```

which stores the full path of where Phenix has been installed. This is useful for 3rd party programs that may want to know the location of Phenix on the PC.

During the installation a command prompt window will appear that compiles all the python sources.

The uninstaller is also defined in mainphenixinstaller.nsi. Uninstalling Phenix is done in 5 steps of which the one lasting the longest is removing all the installed files As a side effect, during uninstallation the progress bar will hang until this step has been completed.

How to set up and use Phenix

install-setup-run.html

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Installation

Chatbot help on installation

You can use the Phenix chatbot to help you with installation.

Platform support

Phenix is supported on most common Linux platforms (glibc 2.17 or newer), macOS version 10.13 or newer for Intel and 11 or newer for Apple Silicon, and Windows 10 or newer.

Space requirements

For the complete Phenix installation you will need approximately 15 GB of disk space. During installation, you will need approximately 25 GB for temporary files.

Graphical installer - Mac

There are installers for Intel processors and for Apple Silicon processors. Choose the installer that matches the hardware on your system. Apple Silicon installers will not run on Intel systems and the Intel version of

Phenix will run significantly slower on Apple Silicon systems.

After downloading the Phenix-xxxx.pkg file use "Open with" and choose "Installer.app". When asked about unknown applications go ahead and hit Open (if you just double-click on the file it will say that it is not approved by Apple and it will not give you the option to approve it yourself). On newer versions of macOS, you will get a message that the installer cannot be run. You must then go to System Preferences, and then Privacy & Security. At the bottom, you will get an option to run the .pkg installer anyway. Click that to run the installer.

Alternatively, you can run xattr -c <GUI installer name>, to skip the security checks by macOS.

You will need administrative privileges to install Phenix into /Applications. For single user installation, the default location is in ${HOME}/Applications. The Phenix application is found inside the Phenix-<version> folder in those directories.

Graphical installer - Windows

Run the .exe installer and follow the prompts. Windows may have similar security prompts like in macOS. Click "More info" and then "Run anyway" to run the installer.

The destination folder should be a path without spaces.

There will be a new Start Menu folder containing the Phenix GUI application and an item for starting a command-line prompt with the Phenix environment.

Command-line installer (macOS and Linux)

On macOS and Linux, the command-line installers are packaged as .sh files. To install, run

% bash <installer name>

for interactive installations. For batch mode (skips interactive prompts), run

% bash <installer name> -b -p <prefix>/phenix-<version>

The phenix-<version> is needed to isolate the Phenix installation from <prefix>.

Installation time

Installation of the binary version of Phenix requires no compilation, only the extraction of compressed packages and then the generation of some data files. You will probably have to wait about 15 minutes for the installation to complete, depending on the performance of your installation platform. However, installation may take upwards of 30 minutes or more if you have antivirus software or slow disk speeds.

Running Phenix

If you intend to use the graphical interface and installed Phenix using the macOS or Windows graphical installers, no further configuration is necessary - just double-click the icon for the main Phenix GUI. Instructions below are for users of the command-line installer and/or programs.

Setting up the command-line environment

Once you have successfully installed Phenix, to set up your environment please source the phenix_env file in the Phenix installation directory (for example - replace "<version>" with the actual installed version, such as "2.0-5933"):

% source <prefix>/phenix-<version>/phenix_env.sh for sh users % source <prefix>/phenix-<version>/phenix_env.csh for csh users

% source <prefix>/phenix-<version>/phenix_env.sh for sh users

% source <prefix>/phenix-<version>/phenix_env.csh for csh users

To simplify the set up of Phenix, you can add this step to your shell startup script.

If you used the graphical Mac installer, you can find the files in the phenix-<version> folder in /Applications or ${HOME}/Applications.

The following environmental variables should now be defined (here with example values):

```
PHENIX=&lt;installation location&gt;
PHENIX_PREFIX=&lt;installation location&gt;
PHENIX_VERSION=&lt;version&gt;
```

The $PATH is also updated to put Phenix executables before your other commands.

Documentation

You can find documentation in the Phenix GUI (under the Help menu). Alternatively, you can use a web browser to view the documentation supplied with Phenix, by typing:

% phenix.doc

If this does not work because of browser installation issues then you can point a web browser to the correct location in your Phenix installation (for example):

% firefox ${PHENIX_PREFIX}/share/phenix/doc/index.html

For license information please see the LICENSE file.

For the source of the components see SOURCES.

Help

You can join the Phenix bulletin board and/or view the archives:

```
https://phenix-online.org/mailman3/lists/phenixbb.phenix-online.org/
```

Alternatively you can send email to:

```
help@phenix-online.org (if you'd like to ask us questions or report bugs)
```

User interfaces

Different user interfaces are required depending on the needs of a diverse user community. Most modules in Phenix can be run either through a graphical user interface (GUI) or as command-line programs. For new users (and those without experience with Unix/Linux systems), we recommend using the graphical interface.

The Phenix GUI

For new users and anyone unfamiliar with Unix command lines, we recommend using the graphical interface. To run, simply type this command:

```
% phenix &amp;
```

Please see the other documentation files to get more details about the Phenix GUI.

Command Line Interface

Advanced users (or anyone developing automated pipelines) may prefer to use the command-line interface. This is particularly the case when rapid results are required, such as data quality assessment and twinning analysis, or substructure solution at the synchrotron beam line. Tools that facilitate the ease of use at the early stages of structure solution, such as data analyses (phenix.xtriage), substructure solution (phenix.hyss) and reflection file manipulations such as the generation of a test set, reindexing and merging of data (phenix.reflection_file_converter) are available via simple command line interfaces. Most of the larger programs such as phenix.refine and the AutoSol, AutoBuild, and LigandFit wizards are also available as command-line tools.

To illustrate the command line interface, the command used to run the program that carries out a data quality and twinning analyses is:

phenix.xtriage my_data.sca [options]

Further options can be given on the command line, or can be specified via a parameter file:

phenix.xtriage my_parameters.def

A similar interface is used for macromolecular refinement:

phenix.refine my_model.pdb my_data.mtz

Although SCALEPACK and MTZ formats are indicated in the above example, reflection file formats such as D*TREK, CNS/XPLOR or SHELX can be used, as the format is detected automatically.

Help for all command line applications can be obtained by use of the flag --help :

phenix.refine --help

There are also many other command line tools (described in detail elsewhere in this documentation). You can list them all with

phenix.list

or alternatively,

phenix.list ave

to list all methods that contain the characters ave in their names or descriptions.

Note: all commands have their regular name and name qualified with the version. You can always use the version-qualified name to ensure which version of a command you are using (in case you have multiple versions of Phenix or related applications installed).

The runtime options for most Phenix tools are controlled using a lightweight syntax called PHIL (Python Hierarchial Interface Language), which is designed to work both as command-line arguments or as more verbose parameter files. A basic overview of PHIL is available in the overview of file formats.

Troubleshooting

Problems installing on Linux

The command-line installer extracts some files into the temporary directory and then runs those files. On systems where the temporary directory is mounted to be non-executable, you will have to provide an alternate temporary directory. To do so, prepend the TMPDIR environment variable to the installation command.

TMPDIR=<executable temporary directory> bash <installer name>

General installation instructions for Rosetta for use in Phenix

rosetta_install.html

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- Troubleshooting
- Proxy Server Setup

Installation

Rosetta is software from the Baker laboratory at the University of Washington https://www.rosettacommons.org. Rosetta is used by a number of modules in Phenix

- phenix.mr_rosetta
- phenix.simple_homology_model.
- phenix.rosetta_refine

phenix.rosetta_refine requires a compilation of Rosetta from source as it requires some programs for the interface between Phenix and Rosetta that are not in the binary distribution. The first three programs will run with a binary installation of Rosetta (or the compiled installation of Rosetta). From a practical point-of-view, making the effort to install the compiled version of Rosetta means that you can run all phenix Rosetta dependent programs.

Once you have installed Rosetta you need to set the environmental variable $PHENIX_ROSETTA_PATH.

Installation - Using standard Phenix installation

To use the latest Rosetta version, Phenix needs to be compiled using a compiler that is c++11 compatible. This is now the case from Phenix 1.14 and beyond. Users of macOS should use the command-line installer and install XCode to get the compilers.

Currently, there is an issue with using phenix.rosetta_refine and macOS High Sierra (10.13). The rosetta.run_tests step in the installation instructions is generating large files and using large amounts of memory.

Proceed with the following.

- Go to Rosetta Commons website and follow instructions there to get a license and then to download.
- You will need Rosetta version 3.10 or later. It is quicker to download the source bundle. The binaries are not required when installing for ALL programs.
- Make the directory where you want to install Rosetta (if it does not already exist). Move the downloaded file (something like rosetta_src_2019.??.xxxxx_bundle.tgz) there. Then unpack the bundle...

```
% tar xzf rosetta_src_20??.??.?????_bundle.tgz (changing the names to
 match your bundle)
```

- You should have a directory like rosetta_src_2019.??.xxxxx_bundle that contains:

```
demos main tools
```

- The directory containing "main demo tools" is to be called "PHENIX_ROSETTA_PATH". Now you can now set a local environmental variable in your ".profile" (sh or bash shell) or ".cshrc" (c-shell) to mark where rosetta is located: if you are using the bash or sh shells:

```
% export PHENIX_ROSETTA_PATH=/your-path-to-rosetta-here/
```

```
or sh (C-shell):
```

```
% setenv PHENIX_ROSETTA_PATH /your-path-to-rosetta-here/
```

- The final step is to build the interface for rosetta_refine. It will also build Rosetta itself if it is not done yet. Supply as many nproc as available on your machine. If you installed Phenix in a location that requires administrative privileges (e.g. /usr/local on linux or /Applications on macOS), you will also need administrative privileges for building. NOTE: On linux, if you are using GCC 5.1 or later (Ubuntu 16.04 or later), there is an ABI change that requires an additional flag to allow linking between our CentOS 6 binary build and Rosetta. To apply this flag, copy this site.settings file to ${PHENIX_ROSETTA_PATH}/main/source/tools/build/site.settings before running this final step.

```
% rosetta.build_phenix_interface nproc=2
```

You should now be completely ready to go with Rosetta in Phenix that can be tested using

```
% rosetta.run_tests
```

Troubleshooting

See FULL INSTRUCTIONS at https://www.rosettacommons.org/support Build may also be called with ./scons.py bin mode=release -j4 Build may fail because it is unable to find the g++ requested by build. This can be fixed in two ways (1) If you have the correct version of g++ installed, but it is not called by the name expected by the build, simply set a soft link from g++ to e.g. g++-4.6 (2) See: http://morganbye.net/blog/2011/05/rosetta-32-ubuntu-1104 for how to edit to tools/build/basic.settings and tools/build/options.settings and using "scons bin mode=release cxx=gcc cxx_ver=4.5" Build may fail due to shallow warnings. Edit tools/build/basic.settings to comment out all "warn" flags for your operating system.

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Proxy Server Setup

To use the fragment server and your machine is behind a firewall and there is a proxy server you need to go through, then if you use a .hhr file to download files from the PDB then you will need to specify your proxy server. You can use the following command to specify the proxy server (replacing it with YOUR proxy server).

If you are using the bash or sh shells:

```
% export HTTP_PROXY=proxyout.mydomain.edu:8080
```

or sh (C-shell):

```
% setenv HTTP_PROXY proxyout.mydomain.edu:8080
```

Secondary structure restraints

secondary_structure.html

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At lower resolution (lower than ~3A) refinement of biological macromolecules may be unstable. In some cases even secondary structure (SS) elements may be distorted during coordinate refinement. SS elements include helices and sheets in protein structures, base-pairs and stacking bases in nucleic acids. This applies to refinement against both X-ray and EM data. To preserve correct geometry of SS it may be useful to use specifically designed secondary structure restraints. Using SS restraints in moderate-to-low resolution refinement may help to preserve correct geometry of SS elements in the structure and sometimes correct slightly distorted SS elements.

General description

SS restraints are available in most refinement tools in Phenix package. Their behaviour is regulated by the same scope of parameters (see the bottom of the page for the full listing). They could be turned on in GUI by checking "Use secondary structure restraints" checkbox. This is equivalent to command-line option secondary_structure.enabled=True. In this case SS elements will be found automatically and outputted to the resulting .pdb and log files. We recommend to check carefully automatic assignments of SS elements because incorrect assignments can lead to wrong refinement results.

Video Tutorial

The tutorial video is available on the Phenix YouTube channel and covers the following topics:

- Basic overview of secondary structure restraints
- How to use secondary restraints in the phenix.refine and phenix.real_space_refine GUI
- How to perform secondary structure annotation in the GUI

Proteins

Secondary structure elements of proteins are helices and sheets. The main stabilizing factor is their hydrogen bonds.

The PDB format supports 10 different types of helix, but only three of these (alpha, pi, and 3_10) are common in naturally occurring proteins and have easily deciphered hydrogen-bonding rules. User-specified helices default to alpha form (hydrogen bonds between residue n and residue n+4). Sheets may be either parallel or antiparallel.

The most common way to specify secondary structure is to use HELIX and SHEET records in the beginning of .pdb file. There are several ways to obtain such records. Examples of such records:

```
HELIX 1 HA GLY A 86 GLY A 94 1 9
SHEET 1 A 5 THR A 107 ARG A 110 0
SHEET 2 A 5 ILE A 96 THR A 99 -1 N LYS A 98 O THR A 107
SHEET 3 A 5 ARG A 87 SER A 91 -1 N LEU A 89 O TYR A 97
SHEET 4 A 5 TRP A 71 ASP A 75 -1 N ALA A 74 O ILE A 88
SHEET 5 A 5 GLY A 52 PHE A 56 -1 N PHE A 56 O TRP A 71
```

Full reference to HELIX and SHEET records PDB format may be found here Hydrogen bond lengths for protein secondary structure are restrained at 2.9A with sigma 0.05A for N-O. Outliers (defined as N-O bonds greater than 3.5A) will be automatically removed, but you may prevent this with secondary_structure.protein.remove_outliers=False. This may be helpful for very poor low-resolution structures, e.g. to force proper structure on a manually built helix, but it will also cause completely spurious bond restraints to be overlooked. Three hydrogen bond angles are also restrained for alpha-helices. The first and the last bond angles are restrained with RMSD of 10 degrees, inner angles restrained with RMSD of 5 degrees. Target values are the following:

- C - O - N: 155
- CA - N - O: 116
- C - N - O: 121

These angle restraints can be turned off by setting secondary_structure.protein.restrain_hbond_angles=False.

How to obtain HELIX/SHEET records

There are various programs available that can produce HELIX/SHEET records suitable for use in refinement. There are several ways to obtain HELIX/SHEET records for particular structure:

- Check your .pdb file. They may be already there
- Run any refinement program from Phenix suite with secondary structure restraints turned on and look in the resulting .pdb file
- Use phenix.secondary_structure_restraints command-line utility.
- Use any other third-party program that have an option to export secondary structure definition in PDB format.

It is highly recommended to check manually all SS annotations coming from any source. They may often contain errors. We encourage users to verify all secondary structure annotations carefully with model. The format of HELIX/SHEET records is described here. The most common errors are: - Lack of registration part of SHEET record (second part of SHEET record when sense is not 0) - Strands are not combined into sheets, i.e. all SHEET records have different - Wrong sense in SHEET record - Not necessary pi-helices containing only 3 residues

Nucleic acids

Secondary structure elements for nucleic acids are base-pairs and stacking pairs. Base-pairs are supposed to be somewhat planar and stacking nucleobases are supposed to be parallel to each other. Several types of restraints to make these features present in resulting model are available in Phenix. None of them are symmetry-aware, e.g. for PDB ID 9dna basepair hydrogen bonds could not be determined automatically. Nevertheless, they still could be specified manually via geometry_restraints.edits mechanism.

Base-pairs

Restraints to keep correct geometry of base-pairs involves hydrogen bonding, planarity and paralleltiy restraints. Hydrogen bond restraints include distance and angle restraints. Two types of restraints available to keep base-pairs planar: standard planarity restraints and parallelity restraint. We use parallelity by default. User can change it to planarity or even use both of them (not recommended) separately for each defined base-pair in parameter file. User also can change type of base-pairing using Saenger classes*:

The 28 possible base-pairs for A, G, U(T), and C involving at least two (cyclic) hydrogen bonds. Hydrogen and nitrogen atoms displayed as small and large filled circles, oxygen atoms as open circles, and glycosyl bonds and thick lines with R indicating ribose C1' atom. Base-pairs are boxed according to composition and symmetry, consisting of only purine, only pyrimidine, or mixed purine/pyrimidine pairs and asymmetric or symmetric base-pairs. Symmetry elements and are twofold rotation axes vertical to and within the plane of the paper. In the Watson-Crick base-pairs XIX and XX and in base-pairs VIII and XVIII, pseudosymmetry relating only glycosyl links but not individual base atoms is observed. Drawn after compilations in (33,457).

- Saenger (1984), Principles of Nucliec Acid Structure, 120-121. Springer-Verlag New York Inc., New York.

Stacking pairs

We use parallelity restraint to keep stacked pairs parallel to each other.

How to obtain annotations

The annotation could be done automatically during refinement and it will be printed out to .def file for future runs. Another way is to use phenix.secondary_structure_restraints command-line utility which can produce annotations in suitable format. PDB does not have suitable records, therefore the only way now to supply custom annotations is to use Phenix parameter files.

Graphical editor

Documentation coming soon!

References

Programming new geometry restraints: parallelity of atomic groups. O.V. Sobolev, P.V. Afonine, P.D. Adams, and A. Urzhumtsev. J. Appl. Cryst. 48, 1130-1141 (2015).

- Programming new geometry restraints: parallelity of atomic groups. O.V. Sobolev, P.V. Afonine, P.D. Adams, and A. Urzhumtsev. J. Appl. Cryst. 48, 1130-1141 (2015).

List of all available keywords

secondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027

• secondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for

proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027

• ss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.

• from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.

• max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identification

• use_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.

• max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chains

• enabled = False Turn on secondary structure restraints (main switch)

• proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None

- enabled = True Turn on secondary structure restraints for protein
- search_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.
- distance_ideal_n_o = 2.9 Target length for N-O hydrogen bond
- distance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be established
- remove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be established
- restrain_hbond_angles = True
- helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None
- serial_number = None
- helix_identifier = None
- enabled = True Restrain this particular helix
- selection = None
- helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.
- sigma = 0.05
- slack = 0
- top_out = False
- angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current =

None bond_start_previous = None hbonddonor = None acceptor = None

• enabled = True Restrain this particular sheet

• first_strand = None

• sheet_id = None

• sigma = 0.05

• slack = 0

• top_out = False

• angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.

• angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value

• strandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None

• selection = None

• sense = parallel antiparallel *unknown

• bond_start_current = None

• bond_start_previous = None

• hbonddonor = None acceptor = None

• donor = None

• acceptor = None

• nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027

• enabled = True Turn on secondary structure restraints for nucleic acids

• hbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be established

• scale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.

• base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335

- enabled = True Restraint this particular base-pair
- base1 = None Selection string selecting at least one atom in the desired residue
- base2 = None Selection string selecting at least one atom in the desired residue
- saenger_class = 0 Type of base-pairing
- restrain_planarity = False Apply planarity restraint to this base-pair
- planarity_sigma = 0.176
- restrain_hbonds = True Restrain hydrogen bonds length for this base-pair
- restrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pair
- restrain_parallelity = True Apply parallelity restraint to this base-pair
- parallelity_target = 0
- parallelity_sigma = 0.0335
- stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027
- enabled = True Restraint this particular base-pair
- base1 = None Selection string selecting at least one atom in the desired residue
- base2 = None Selection string selecting at least one atom in the desired residue
- angle = 0
- sigma = 0.027